

Plasma proteomics for prognostic stratification

in newly diagnosed primary central nervous system lymphoma

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LIX, École Polytechnique · French LOC network

On behalf of the French LOC network and our collaborators

I have no commercial disclosure



PCNSL: aggressive disease, variable outcomes

Aggressive extranodal lymphoma

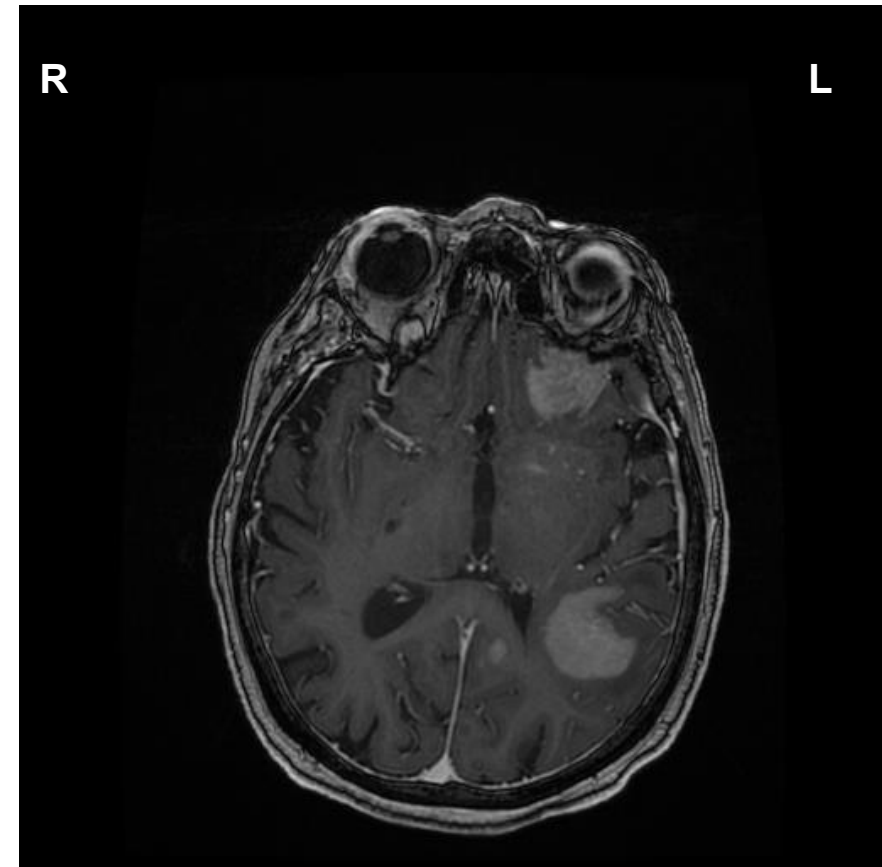
Usually diffuse large B-cell lymphoma confined to the CNS at diagnosis.

Treatment

High-dose methotrexate-based induction.

Heterogeneous outcomes

In the LOC analysis cohort (n = 162):
83 patients had progression or death before 9 months (51%).



Axial T1-weighted MRI after contrast

Can plasma proteins add to clinical risk assessment?



Clinical assessment

Age · performance status · sex

MSKCC-like classes (analysis)

- | | |
|---|--------------------|
| 1 | Age < 50 years |
| 2 | Age ≥ 50; KPS ≥ 70 |
| 3 | Age ≥ 50; KPS < 70 |



Plasma-protein information

Does the baseline immune proteome add prognostic information?

Prespecified endpoints

Overall survival (OS)

Progression-free survival (PFS)

Progression or death before 9 months

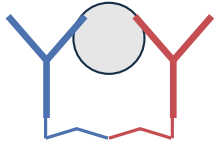
Clinical prediction baseline: age ≥ 60, KPS ≥ 70 and sex. Cox association models adjust for MSKCC class and sex.

Develop in earlier patients. Test in later patients.

 LOC network · 162 patients · Plasma at diagnosis · 1007 measured proteins

Proximity extension assay: protein binding → DNA reporter → abundance

1 Dual protein recognition



Two antibodies carrying DNA tags

2 DNA extension



Proximity-dependent polymerisation

3 Reporter quantification



PCR → sequencing → NPX ($\log_2$)

Targets: cytokines / chemokines (IL6, TNF, CXCL9), receptors (PDCD1, LAG3), signalling (ABL1, PTEN)

Training: n = 84

Jan 2016 – Jun 2019

Define clusters, cut-offs and models

Temporal test: n = 78

Jul 2019 – Dec 2022

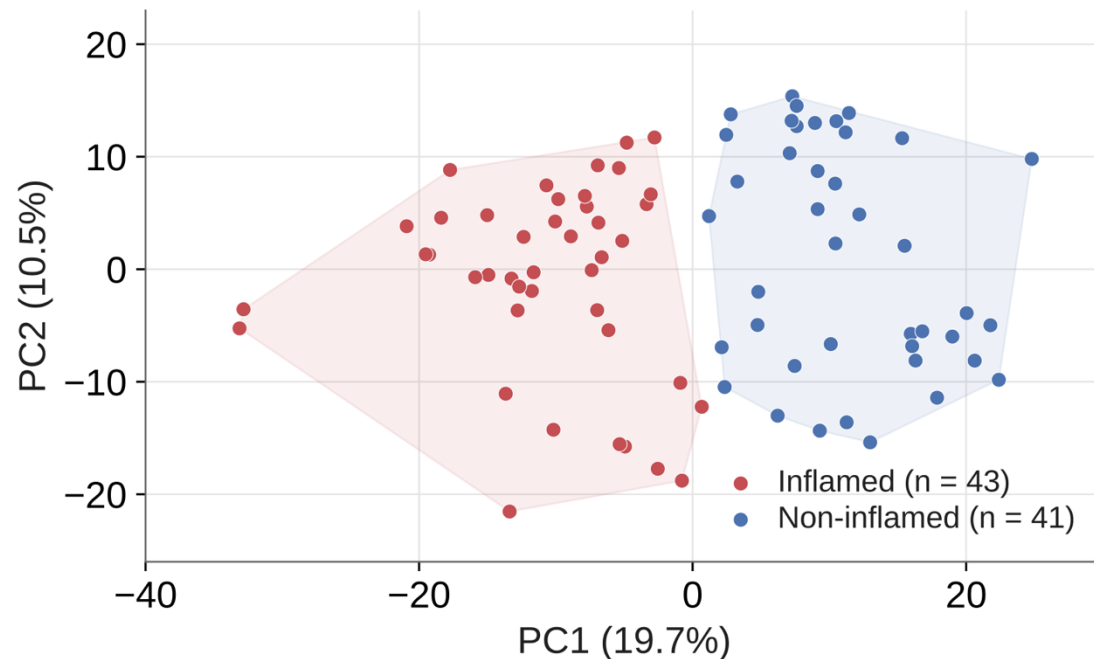
Apply the fitted approach unchanged

NPX: relative protein abundance · Training: 86 enrolled, 2 technical exclusions · Replicates: mean $r = 0.955$ (8 pairs).

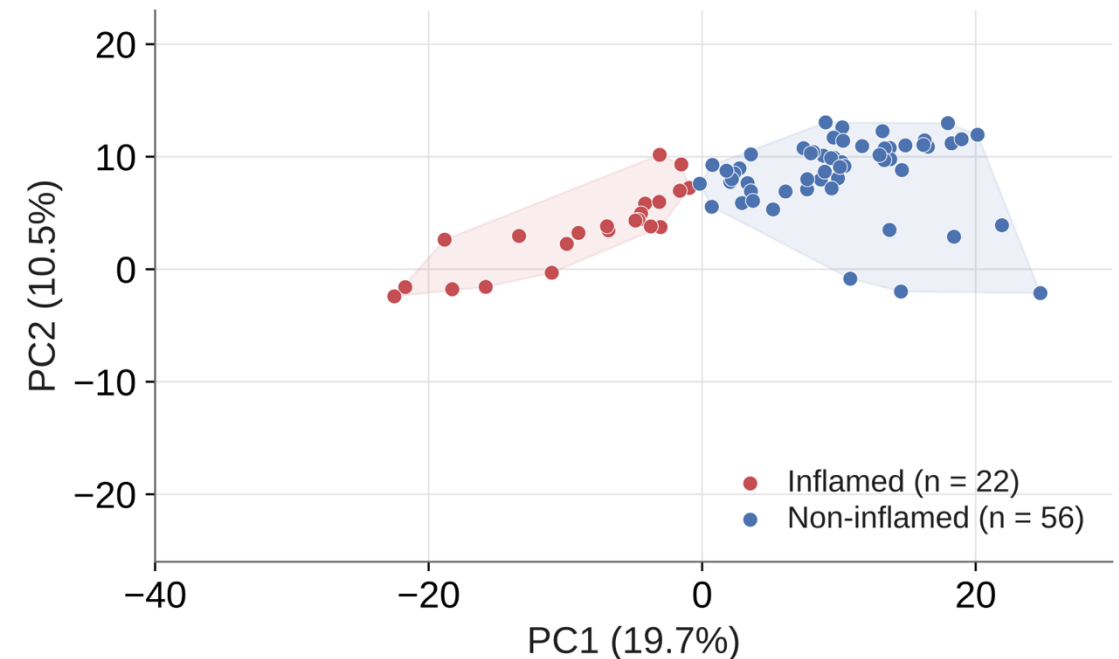
Two plasma profiles recur in later patients

Consensus k-means: 1007 proteins; 500 resamples; 80% of training patients per resample; $k = 2$.

a Training (n = 84)



b Temporal test (n = 78)



211 / 267 training differential proteins replicate in test with the same direction (79%).

Welch tests; BH-FDR < 0.05 and $|\Delta\text{NPX}| > 1$. Test projected into training PCA space; hulls are not confidence regions.

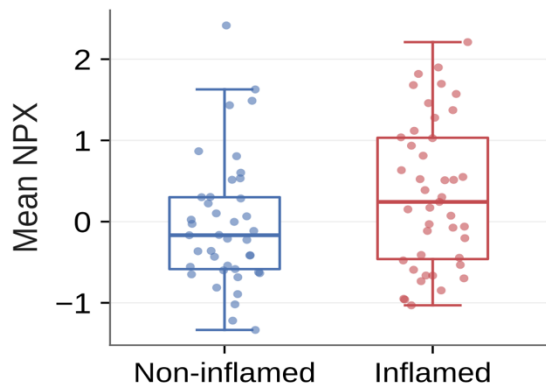
Four signatures describe the immune profiles

Training n = 84 · Focus: seven-protein innate score; training median defines high / low.

IFN / chemokine

7 proteins

Cohen's d = 0.44

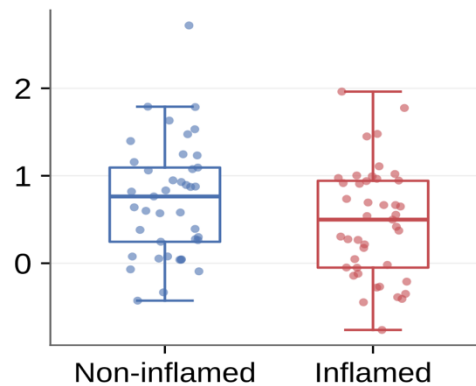


CXCL9 · CXCL10 · IFNG
CCL19 · CCL20 · CCL3 · CCL4

Checkpoint / cytokine

7 proteins

Cohen's d = -0.40

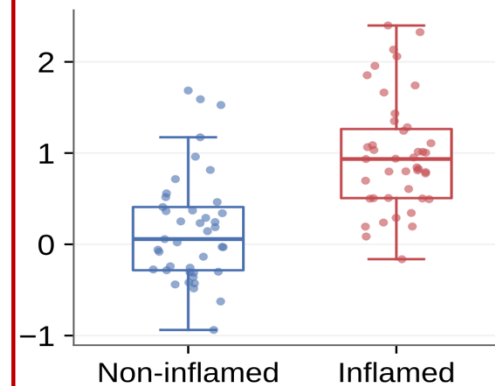


IL10 · IL6 · PDCD1 · PDCD1LG2
LAG3 · CD86 · TNF

Innate / inflammatory

7 proteins

Cohen's d = 1.34

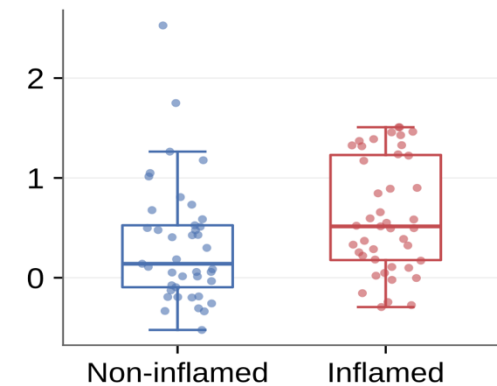


IL1B · IL6 · TNF · CCL20
CCL3 · CCL4 · FASLG

Integrated immune

17 proteins

Cohen's d = 0.49



IL10 · IL6 · IL1B · CCL19
PDCD1LG2 · PDCD1 · CXCL9
CXCL10 · IFNG · LAG3 · CD86
TNF · CCL20 · CCL3 · CCL4
TNFRSF13C · FASLG

Descriptive comparison; hypergeometric enrichment: all BH-adjusted P = 1.0. Overlapping panels contain 17 distinct proteins.

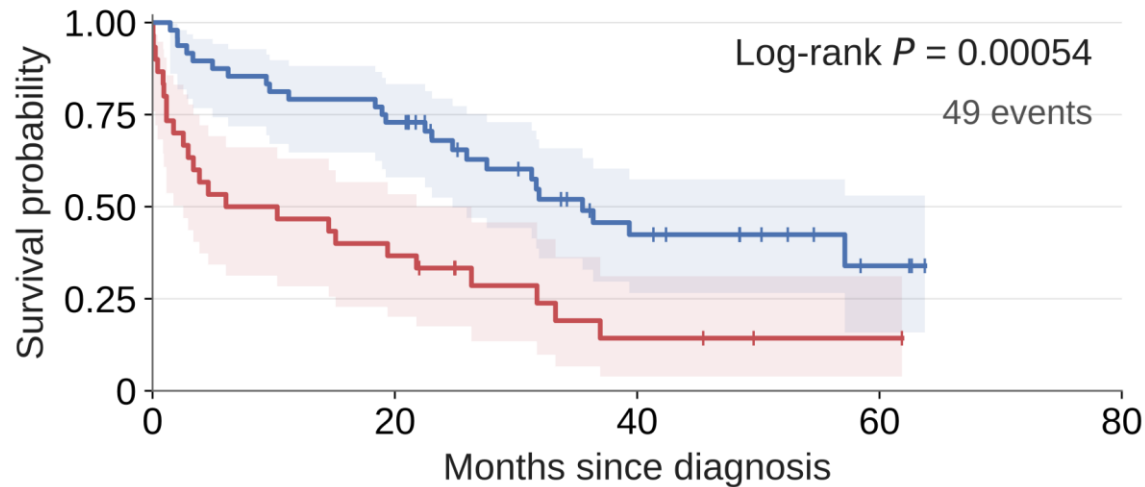
Higher innate scores are associated with shorter survival

Seven proteins: IL1B · IL6 · TNF · CCL3 · CCL4 · CCL20 · FASLG

Temporal test cohort: n = 78

Score = mean NPX; high ≥ 0.5037 (training median; applied unchanged to test).

a Overall survival



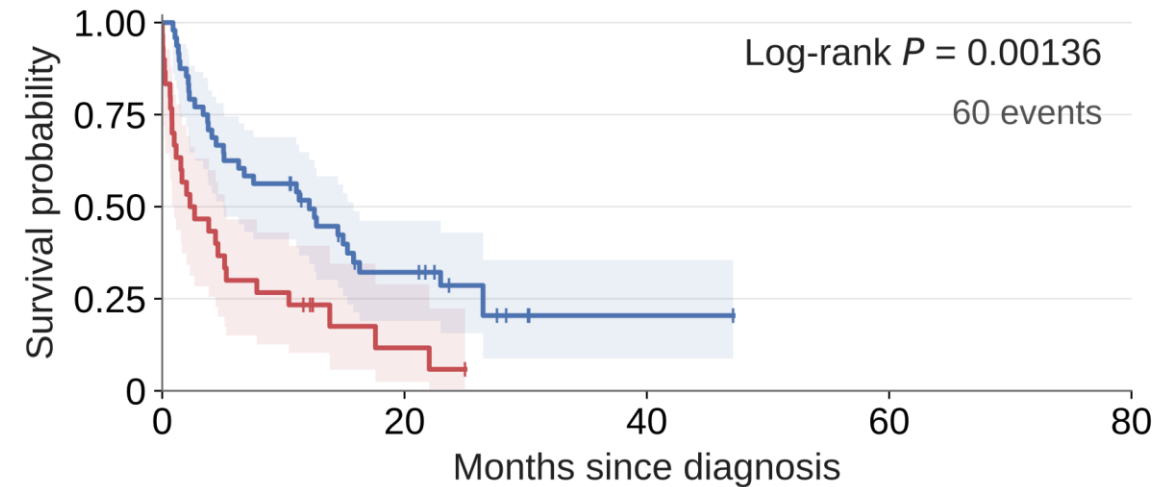
Number at risk

High	30	11	3	1	0
Low	48	35	13	3	0

Adjusted HR 2.51 (95% CI 1.39–4.52); $P = 0.0022$

Cox: high vs low, adjusted for MSKCC class and sex (Wald P). Shading: 95% CI; ticks: censoring.

b Progression-free survival



Number at risk

High	30	2	0	0	0
Low	48	12	1	0	0

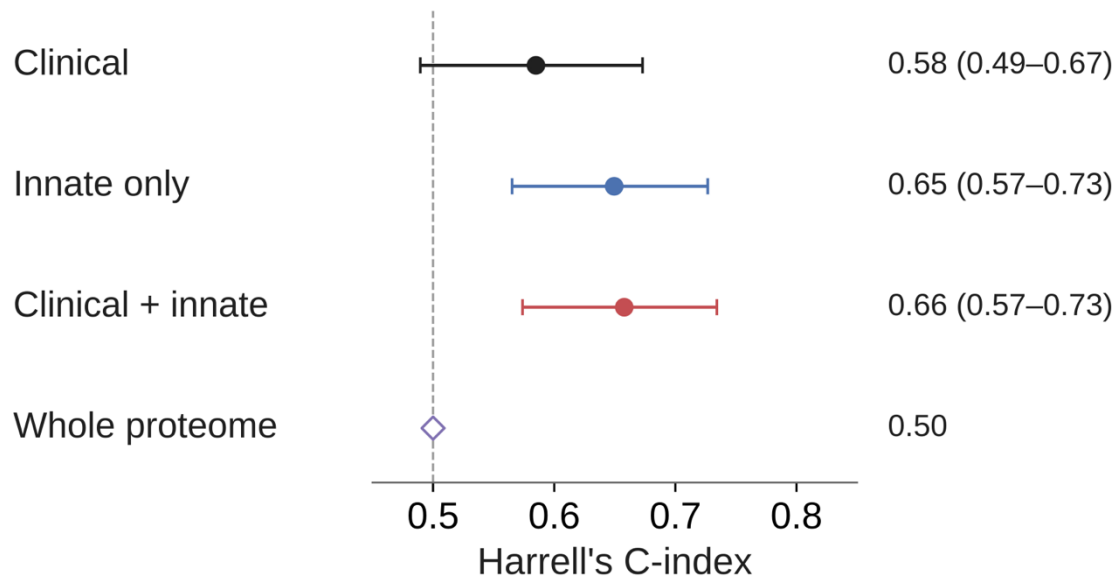
Adjusted HR 2.05 (95% CI 1.19–3.55); $P = 0.0097$

Innate score adds discrimination to clinical variables

Innate score: IL1B · IL6 · TNF · CCL3 · CCL4 · CCL20 · FASLG

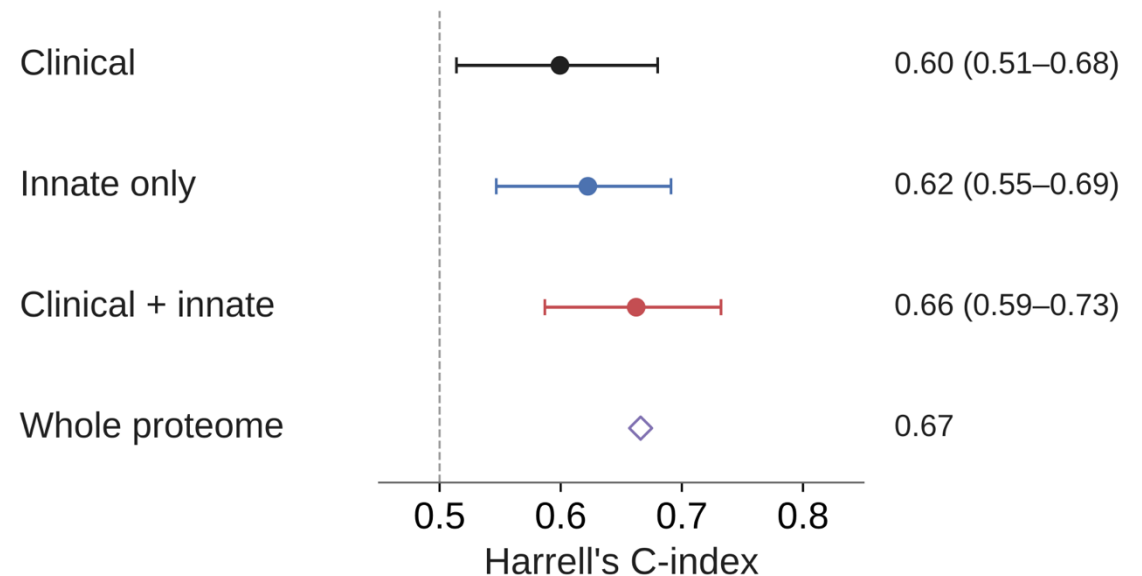
Cox prediction models fitted in training (n = 84), evaluated in temporal test (n = 78).

a Overall survival



ΔC , combined – clinical: +0.073 (0.004–0.151)

b Progression-free survival



ΔC , combined – clinical: +0.063 (0.001–0.131)

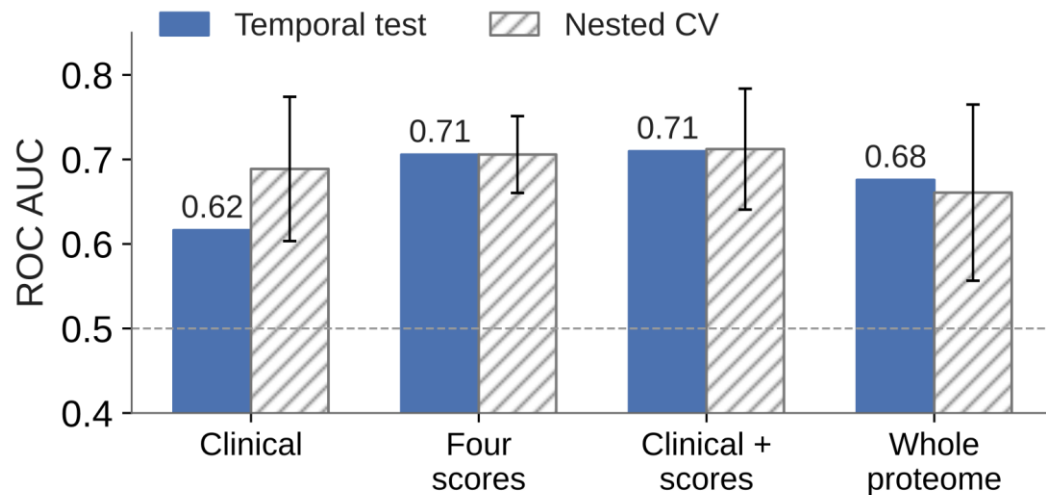
95% CI: 5,000 fixed-model test bootstraps. Whole-proteome points have no interval. Clinical: age $\geq$ 60, KPS $\geq$ 70, sex.

Immune scores outperform clinical variables for early failure

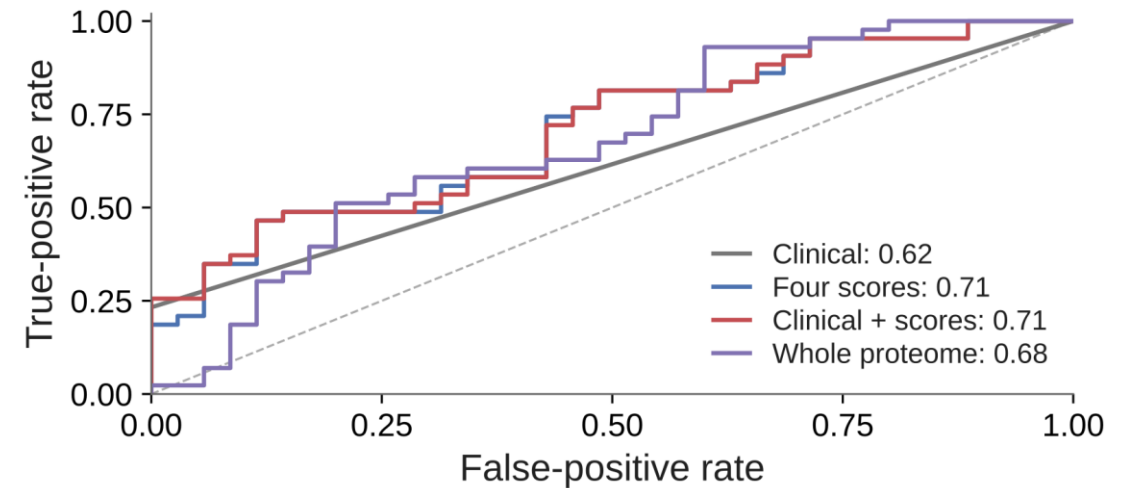
Early failure: PFS event < 9 months · Elastic-net logistic regression

Evaluable: training n = 80 / test n = 78; 4 early-censored training patients excluded.

a Discrimination



b Temporal test ROC curves



Four scores / 17 proteins: IL10 · IL6 · IL1B · CCL19 · PDCD1LG2 · PDCD1 · CXCL9 · CXCL10 · IFNG
LAG3 · CD86 · TNF · CCL20 · CCL3 · CCL4 · TNFRSF13C · FASLG

Δ AUC, combined – clinical: +0.093 (95% CI –0.012 to 0.197). CV error bars = fold SD; bootstrap CI holds models fixed.

Plasma proteins may refine risk assessment

LOC cohort · Training n = 84 → Temporal test n = 78

1 Reproducible immune-related plasma profiles

211 / 267 differential proteins replicated in test (79%).

2 Seven-protein innate score associated with survival

Adjusted HR (95% CI): OS 2.51 (1.39–4.52); PFS 2.05 (1.19–3.55).

High vs low score; adjusted for MSKCC class and sex.

IL1B · IL6 · TNF · CCL3 · CCL4 · CCL20 · FASLG

3 Immune scores add prognostic information

OS C-index: 0.58 → 0.66 with clinical + innate score.

Early-failure AUC: 0.71 (95% CI 0.59–0.82), clinical + four scores.

Next: independent validation and clinical evaluation

External cohorts · calibration · prospective assessment

Acknowledgements

We thank the **patients and their families**, the **French LOC network** and the **Onconeurotek v2 biobank**.

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Published in

Neuro-Oncology

Künzle et al., 2026

[doi:10.1093/neuonc/noag190](https://doi.org/10.1093/neuonc/noag190)

FUNDING

LOCImm · BRAINTWIN · MultiPOLA
SiRIC CURAMUS · SOUND.AI
ERC VEGA



Co-funded by
the European Union